

Manifold for T1D – VX-880 Post-Infusion Analytics

Stem-cell-derived islet-therapy graft-function hazard and response decomposition

VX-880's initial cohort achieving **insulin-independence** is a stage-change proof-of-principle for stem-cell-derived islet therapy. The next five years will be defined by analytic questions, not biological ones: who reaches durable engraftment, what precedes graft compromise, and which second-generation regimens and adjuncts (including the encapsulated VX-264 variant) most attenuate loss-of-function hazard. Each post-infusion subject contributes an exceptionally dense longitudinal record — C-peptide AUC, CGM, inflammatory panels, immunosuppression pharmacokinetics, autoantibody titers — whose *joint dynamical structure* is under-exploited by endpoint-by-endpoint biostatistics. **Manifold** configures three engine capabilities to the domain: causal discovery on multi-signal post-infusion data, criticality estimators (Gaussian-process regression, Bayesian change-point, variance-based early warning, HTE learners) feeding a composite fragility score (ΩF), and Pearl do-calculus with interdiction for principled second-generation regimen design.

VX-880's first patients reaching insulin-independence is a proof-of-principle, not a finished product. The outstanding question is no longer whether β -cell function can be restored, but for which patients, with which immunosuppression regimens and adjuncts, and over what duration — a problem of causal inference on small- n high-signal trajectories.

ΩF PILLAR MAPPING – POST-INFUSION PHYSIOLOGY

ΩF PILLAR	VX-880 POST-INFUSION READING
I — Interaction	Donor-cell $\times$ recipient-HLA $\times$ immunosuppression exposure interactions; graft $\times$ host-immune compatibility
R — Reserve	Post-infusion graft functional mass (C-peptide AUC on MMTT); stress-test response
J — Junction	Immune-metabolic coupling: cytokine-panel $\leftrightarrow$ C-peptide transfer entropy; autoantibody $\times$ graft-function
C — Cascade	Early-inflammation cascade post-infusion; allo- and auto-immune escalation triggers
T — Temporal	Hazard of graft compromise, response loss, or insulin-dependence return

PROPOSED PARTNERSHIP

Partnership with a VX-880 trial principal investigator at the participating site is sought for:

1. Retrospective post-infusion trial data access under the site's existing Data Use Agreement.
2. Clinical co-authorship on the graft-function hazard and response-decomposition benchmark manuscript (twelve to fourteen weeks from data access).
3. Co-application to Breakthrough T1D Cures, Helmsley Restoring Insulin Production, or NIH NIDDK for the Phase 2 prospective arm.

Apex Analytica contributes the causal-inference engine, the ΩF framework, and the engineering team. Full brief: M-T1D-02-VX880 .